ITTHYNK SMART SOLUTIONS EVENT MANAGEMENT SYSTEM

Problem Understanding

1. The Problem

The client regularly organizes different types of events, including workshops, training sessions, networking events and company activities. Currently, information about these events and the people attending them is managed using different methods.

As the number of events increases, this approach can become difficult to manage because event information and attendee information may be spread across separate processes or tools. This can make it harder for organizers to maintain accurate event details, manage registrations consistently and quickly determine who is attending a particular event.

The current approach may also create problems such as duplicated information, missing or outdated records, difficulty tracking registrations and limited visibility of attendee numbers.

The client therefore requires a more centralized and digital way of managing event information and registrations.

2. People Affected

Event Organizers

Event organizers are affected because they are responsible for creating events, maintaining event information and managing the people who register.

Using different methods can make it difficult to keep information organized and determine the current status of registrations. Organizers may need to manually combine information from different sources to understand who is attending an event.

Attendees

People interested in attending events may be affected because event information and registration processes may not be centralized.

They need a clear way to view available events, access important event information and register for events.

Organization or Client

The organization is affected because inefficient event management can make it difficult to maintain accurate information as the number of events increases.

The organization needs better visibility of events and attendance so that event activities can be managed more effectively.

3. Proposed Solution

The proposed solution is a centralized, web-based Event Management System developed using React and Next.js.

The system will provide a single platform for managing event information and attendee registrations. Authorized event organizers will be able to create and manage events, while attendees will be able to browse available events and register for them.

The system will also allow organizers to view registration information for each event, providing a clearer understanding of who is attending and how many people have registered.

The first version of the system will focus on solving the client's core problems rather than adding unnecessary features.

4. Success Criteria

The project will be considered successful if the system enables the client to:

- Manage event information from a centralized platform.
- Create, update and manage events more efficiently.
- Provide users with a clear way to view available events.
- Manage attendee registrations digitally.
- View the people registered for each event.
- See the number of registrations associated with an event.
- Reduce the need to manually combine event and attendee information from different methods.
- Maintain more consistent and accessible event records.
- Support different types of events as the organization's event activities grow.

Important planning decision

I recommend that our team **does not immediately add features such as**:

- Online payments
- QR-code check-in
- Email notifications
- Calendar synchronization
- AI recommendations
- Advanced analytics

These may be useful in a future version, but they are **not directly required by the client's brief**. We can mention them as potential future enhancements without making them part of the MVP.

Stakeholders and Users

Based on the client brief, the following stakeholders and users have been identified.

User / Stakeholder	Role	Goal	Main Actions
Event Organiser (An employee responsible for organising workshops, training sessions, networking events, or company activities.)	Organises and manage company events as well as the attendees	Manage events and attendees efficiently • Create events easily. • Keep event information up to date. • View registrations. • Know the expected number of attendees. • Access attendee information.	Create, edit and view events as well as registrations • Create an event. • Edit an event. • Publish an event. • View event details. • View registered attendees. • Close registration when appropriate.
Administrator (A person responsible for overseeing the Event Management System.)	Oversees the system	Maintain accurate system information • Maintain accurate information. • Manage users. • Monitor events. • Control access.	Manage users, events and registrations • Manage user accounts. • Manage event records. • View registrations. • Manage user roles.

			• Access system-level information.
Attendee (A person who wants to participate in one of the client's events)	Participates in events	Find and register for events • Find relevant events. • View event details. • Register successfully. • Keep track of events they have registered for	Browse events, view details, register • Create an account. • Log in. • Browse events. • View event details. • Register for an event. • View registration status.
Management (Managers or decision-makers within the client organisation.)	Oversees organisational activities	Understand event activity and participation • Monitor organisational events. • Understand participation. • Make informed decisions about future events.	View event and registration information • View event information. • View registration statistics. • View participation information.

1.Planning System Requirements for an Event Management System

System requirements define **what the Event Management System must do** and **how the system should perform**.

A. Functional Requirements

1. User Registration and Login

The system should allow users to:

- Create an account.
- Log in securely.
- Log out.
- Reset their password.
- Manage their profile.

2. Event Management

Administrators or event organizers should be able to:

- Create events.
- Edit event information.
- Delete events.
- Publish and unpublish events.
- Set event date, time, location, and description.
- Set the maximum number of attendees.

3. Event Browsing

Users should be able to:

- View available events.
- Search for events.
- Filter events by category, date, or location.
- View detailed event information.

4. Event Registration

Users should be able to:

- Select an event.
- Register for an event.

- Enter their required information.
- Receive confirmation after registration.
- Cancel their registration where permitted.

5. Ticket Management

The system should:

- Generate tickets after successful registration.
- Assign a unique ticket/reference number.
- Display ticket information.
- Allow users to view their tickets.
- Support different ticket types if required.

6. Payment Management

If the event requires payment, the system should:

- Display the ticket price.
- Process payments securely.
- Confirm successful payments.
- Record payment status.
- Provide a receipt or payment confirmation.

7. Attendee Management

Event organizers should be able to:

- View registered attendees.
- Search for attendees.
- Update attendee information.
- Check attendees in.
- View attendance status.

8. Notifications

The system should send notifications for:

- Registration confirmation.
- Payment confirmation.
- Event reminders.
- Event cancellations.
- Event date/time changes.

9. Event Check-In

The system should allow organizers to:

- Verify attendee tickets.
- Mark attendees as checked in.

- Prevent duplicate check-ins.
- View the number of attendees who have checked in.

10. Reports

Administrators should be able to generate reports showing:

- Number of registered attendees.
- Number of attendees who checked in.
- Available tickets.
- Tickets sold.
- Revenue.
- Upcoming and completed events.

B. Non-Functional Requirements

These describe **how the system should perform**.

1. Security

- User information must be protected.
- Passwords must be securely stored.
- Users should only access information they are authorized to view.
- Payment information must be handled securely.

2. Performance

The system should respond quickly when users:

- Search for events.
- Register for events.
- Purchase tickets.
- View their tickets.

3. Availability

The system should be available when users need to register for events, especially during periods of high traffic.

4. Usability

The system should have:

- A simple interface.
- Clear navigation.
- Easy registration.

- Mobile-friendly pages.

5. Reliability

The system should accurately record:

- Registrations.
- Payments.
- Tickets.
- Attendance.
- Event information.

6. Scalability

The system should be able to support an increasing number of:

- Users.
- Events.
- Registrations.
- Tickets.

7. Maintainability

Developers should be able to easily:

- Fix bugs.
- Update the system.
- Add new features.
- Modify event functionality.

8.Assumption

The Event Management System assumes that users have internet access and valid accounts, event organizers provide accurate event information, events may be free or paid, registration is required for attendance, and authorized administrators manage users and events. The system is assumed to use a reliable database, secure authentication, and a secure payment provider where applicable.

C. Main Users

User	Responsibilities
Attendee	Browse events, register, pay, and manage tickets

User	Responsibilities
Event Organizer	Create events, manage attendees, and check people in
Administrator	Manage users, events, payments, and reports

D. Basic System Flow

```
User
  ↓
Register / Login
  ↓
Browse Events
  ↓
Select Event
  ↓
Register
  ↓
Payment (if required)
  ↓
Ticket Generated
  ↓
Event Reminder
  ↓
Check-In
  ↓
Attendance Recorded
  ↓
Reports
```

Technology Stack to be utalized

For a web-based Event Management System, you could use:

- **Frontend:** React.js
- **Backend:** Node.js
- **Database:** MySQL, PostgreSQL, or MongoDB
- **Authentication:** JWT / session authentication
- **API:** Postman
- **Version Control:** Git and GitHub
- **Hosting:** Azure, AWS, Afrihost or another cloud platform

Main objective: Build a secure and user-friendly system that allows organizers to create and manage events while allowing attendees to discover events, register, receive tickets, and attend events.

4. USER STORIES AND ACCEPTANCE CRITERIA – EVENT MANAGEMENT SYSTEM

STORY 1: View Available Events

As a customer (attendee), I want to view available event times, So that I can choose a suitable time.

Acceptance Criteria:

- Given that a customer is viewing available event times
- When they filter by date
- Then only events with available seats for that date should be shown.

STORY 2: Book an Event (Main Story)

As a customer, I want to select an available time and complete the booking, So that my seat is secured.

Acceptance Criteria:

- Given that a customer is viewing available appointment times
- When they select an available time and successfully complete the booking
- Then the appointment should be recorded and the selected time should no longer be available and a QR ticket should be generated.

STORY 3: Create Event – Organizer

As an organizer, I want to create a new event with title, venue, date, capacity, So that customers can book it.

Acceptance Criteria:

- Given that I am logged in as organizer

- When I submit valid event details
- Then event should be created as PENDING and sent to admin.

STORY 4: Approve Event – Admin

As an admin, I want to approve events, So that only verified events go live.

Acceptance Criteria:

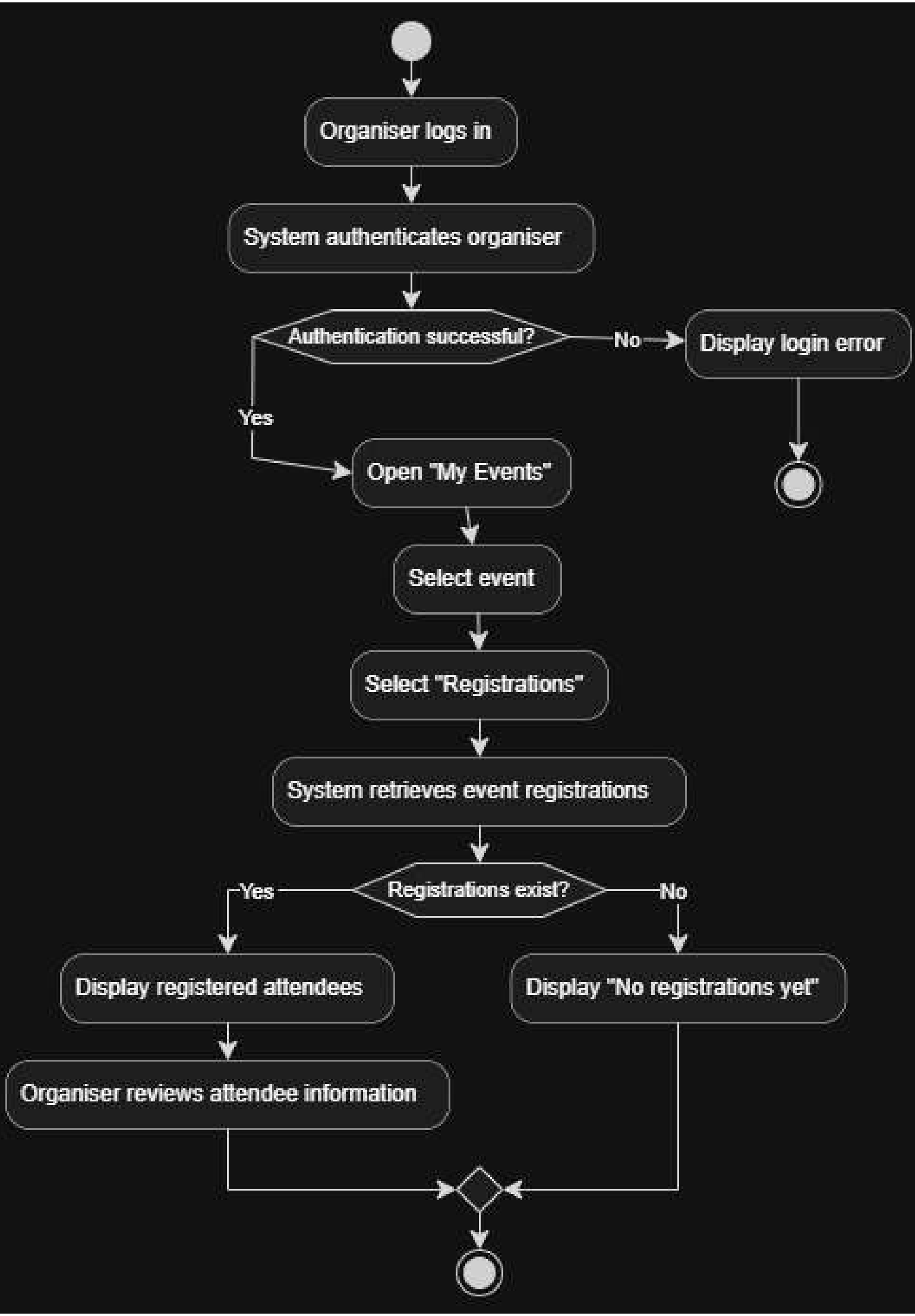
- Given that an event is PENDING
- When admin approves it
- Then event becomes PUBLISHED and visible to customers.

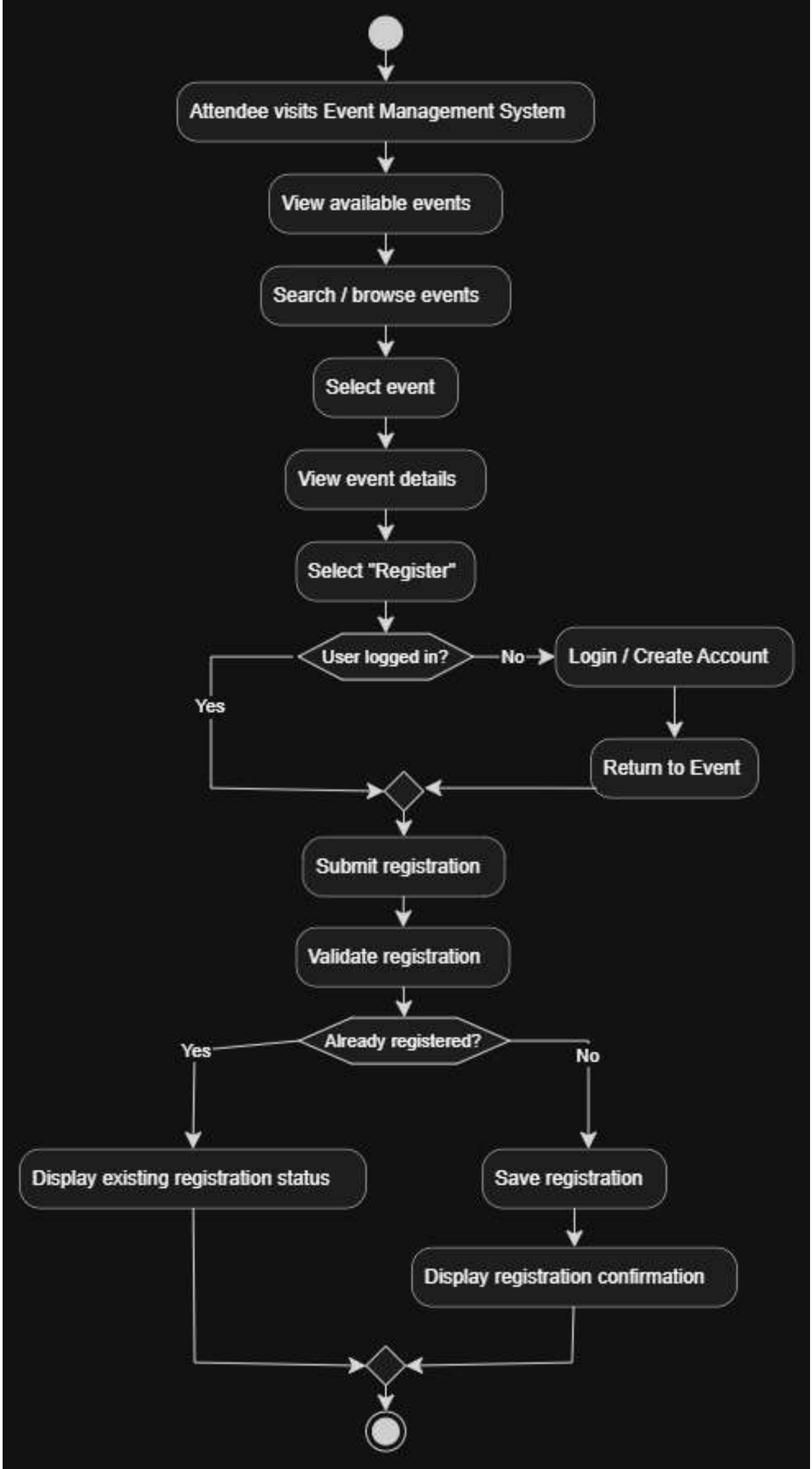
STORY 5: Check-in at Venue

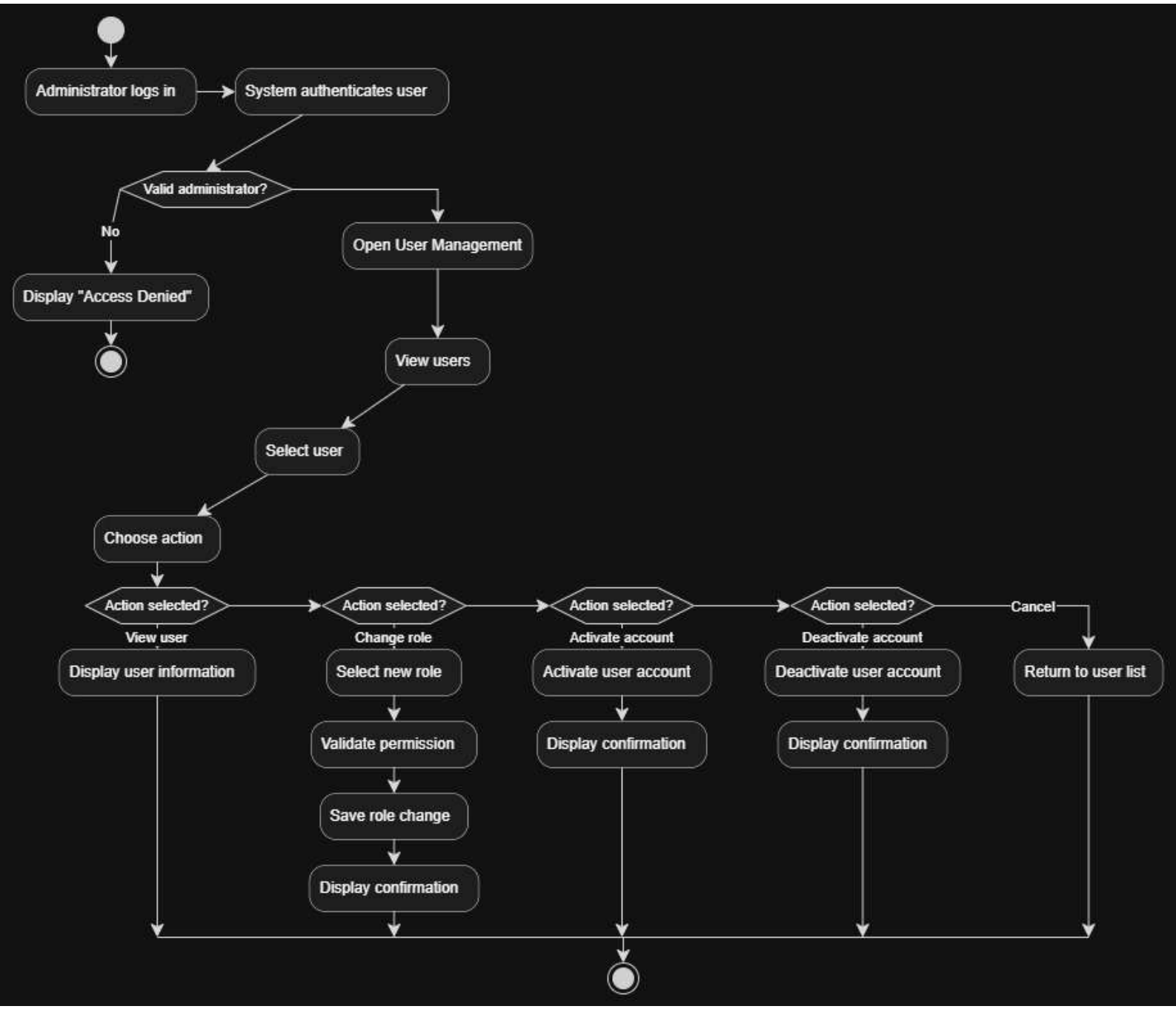
As an organizer, I want to scan customer QR tickets, So that I can verify valid entry.

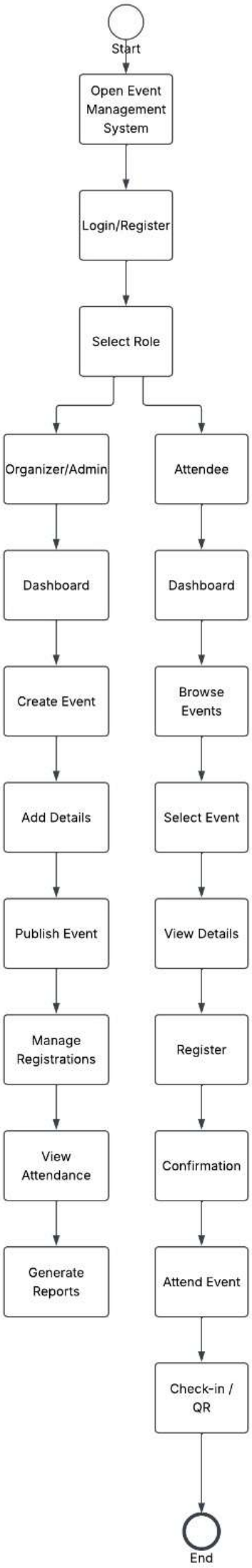
Acceptance Criteria:

- Given that customer shows valid QR
- When organizer scans it
- Then ticket should be marked USED and entry allowed.


Organiser logs in
System authenticates organiser
Authentication successful?
No
Display login error
Yes
Open "My Events"
Select event
Select "Registrations"
System retrieves event registrations
Registrations exist?
Yes
No
Display registered attendees
Display "No registrations yet"
Organiser reviews attendee information


Attendee visits Event Management System
View available events
Search / browse events
Select event
View event details
Select "Register"
User logged in?
No
Login / Create Account
Yes
Return to Event
Submit registration
Validate registration
Already registered?
Yes
No
Display existing registration status
Save registration
Display registration confirmation


Administrator logs in
System authenticates user
Valid administrator?
No
Display "Access Denied"
Open User Management
View users
Select user
Choose action
Action selected?
View user
Display user information
Action selected?
Change role
Select new role
Validate permission
Save role change
Display confirmation
Action selected?
Activate account
Activate user account
Display confirmation
Action selected?
Deactivate account
Deactivate user account
Display confirmation
Cancel
Return to user list


Start
Open Event Management System
Login/Register
Select Role
Organizer/Admin
Attendee
Dashboard
Dashboard
Create Event
Browse Events
Add Details
Select Event
Publish Event
View Details
Manage Registrations
Register
View Attendance
Confirmation
Generate Reports
Attend Event
Check-in / QR
End